

How DICE-2023 Includes Climate Tipping Points

The Short Answer

DICE-2023 includes climate tipping points as a **separate additive component** in the damage function, adding an extra **1% GDP loss at 3°C warming**. This is based on a comprehensive 2021 study by Dietz et al. that modeled the economic impacts of eight major climate tipping points.

Importantly: The tipping points are NOT modeled dynamically or probabilistically. Instead, they're incorporated as a **static adjustment** to the smooth quadratic damage function.

The Three Components of the DICE-2023 Damage Function

The total damages in DICE-2023 are built from **three separate pieces**:

Component 1: Literature Synthesis of Direct Impacts

- **1.62% GDP loss at 3°C**
- Based on 56 estimates from 33 published studies
- Includes things like agricultural losses, health impacts, sea level rise, energy costs
- Updated from 1.2% in DICE-2016

Component 2: Tipping Points Adjustment ← THIS IS THE KEY

- **+1.0% GDP loss at 3°C**
- Based entirely on Dietz et al. (2021) study
- Represents the economic consequences of crossing major climate thresholds

Component 3: Judgmental Adjustment for Omitted Impacts

- **+0.5% GDP loss at 3°C**
- Covers things we know are missing: biodiversity loss, ocean acidification, extreme events, social unrest, migration
- A "fudge factor" for humility about what we don't know

TOTAL: 3.1% GDP loss at 3°C warming

All three components use the **same quadratic functional form**: proportional to the square of temperature change from preindustrial levels (1765).

What Tipping Points Are Included?

The Dietz et al. (2021) study modeled **eight major tipping elements**:

1. **Greenland Ice Sheet** - Complete melting over centuries
2. **West Antarctic Ice Sheet** - Collapse of ice shelves
3. **Atlantic Meridional Overturning Circulation (AMOC)** - Gulf Stream shutdown
4. **Amazon Rainforest** - Dieback to savanna
5. **Boreal Forests** - Northern forest dieback
6. **Permafrost** - Methane and CO₂ release from thawing
7. **Arctic Sea Ice** - Summer ice loss
8. **Coral Reefs** - Mass die-off

Each of these has:

- A **threshold temperature** where it becomes likely to tip
- A **time scale** over which the tipping unfolds
- **Economic consequences** from the resulting changes

How the Dietz et al. Model Works

The Dietz paper is sophisticated - they created a "meta-model" that:

1. **Assigned probabilities** to each tipping point occurring at different temperature levels
 - Some tips are very likely at 2°C (coral reefs)
 - Others need 3-4°C (AMOC, ice sheets)
2. **Modeled the economic impacts** of each tip:
 - Sea level rise from ice sheets → coastal flooding
 - AMOC slowdown → European cooling, precipitation changes
 - Amazon dieback → carbon release, regional climate changes
 - Permafrost thaw → additional greenhouse gas emissions

3. **Ran 10,000 Monte Carlo simulations** to capture uncertainty

4. **Calculated the expected economic loss** across all scenarios

Their central estimate: At 3°C warming, tipping points add about **25% more damage** on top of the smooth impacts we already expect.

How DICE-2023 Simplifies This

Here's the key simplification: Instead of incorporating all that complexity, DICE-2023 takes Dietz et al.'s **bottom-line number** and adds it as a simple adjustment.

The Mathematical Implementation:

The damage function is:

$$\Omega(t) = \psi_1 \times \text{TAT}(t) + \psi_2 \times [\text{TAT}(t)]^2$$

Where:

- $\Omega(t)$ = Fraction of GDP lost to climate damages
- $\text{TAT}(t)$ = Temperature increase since 1765 (°C)
- ψ_1 = Linear coefficient
- ψ_2 = Quadratic coefficient

To include tipping points, they just adjusted the coefficients ψ_1 and ψ_2 so that:

- At 3°C: Total damage = 3.1% (not 2.12% without tipping points)
- Maintains the quadratic form

In Plain English:

They took the tipping points research and "baked it into" the smooth damage curve. The curve is just **steeper** than it would be without tipping points.

What this means:

- At 1°C warming: Damages are slightly higher than without tipping points
- At 2°C warming: Damages are moderately higher
- At 3°C warming: Damages are 1% of GDP higher

- At 4°C warming: The adjustment scales up quadratically
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What's NOT Modeled

This approach **does not** capture several important features of real tipping points:

1. No Sudden Jumps

Real tipping points are **threshold effects** - nothing happens until you cross a threshold, then BAM, big changes.

DICE-2023's smooth quadratic function means damages increase **gradually** as temperature rises. There's no sudden spike when we hit 2.5°C or 3°C or whatever.

2. No Irreversibility

Real tipping points often have **hysteresis** - once you tip, you can't untip by just lowering temperature back below the threshold.

DICE-2023's damage function is **reversible** - if temperature goes back down, damages go back down proportionally.

3. No Probability Distributions

Dietz et al. modeled tipping points **probabilistically** - at 2.5°C there's maybe a 40% chance of Amazon dieback, at 3°C it's 60%, etc.

DICE-2023 uses the **expected value** - a single number. No fat tails, no small probability of catastrophe.

4. No Dynamic Feedbacks

Some tipping points cause **additional warming** (permafrost releases methane, Amazon stops absorbing CO₂).

DICE-2023 doesn't capture this - the climate model and damage model are separate. Tipping points don't make the temperature rise faster.

5. No Individual Thresholds

Different tipping points happen at different temperatures. Coral reefs are dying now at 1.5°C. Ice sheets might not collapse until 3-4°C.

DICE-2023 treats it as one smooth adjustment. You don't see coral reef damage at 1.5°C followed by ice sheet damage at 3°C.

Why This Approach?

Advantages:

1. **Simple** - Keeps the model tractable for optimization
2. **Conservative** - Includes tipping points rather than ignoring them
3. **Based on peer-reviewed research** - Not just speculation
4. **Maintains model structure** - Doesn't require wholesale redesign

Disadvantages:

1. **Understates catastrophic risk** - Smooth curves don't capture "oh shit" moments
 2. **Hides uncertainty** - Uses expected value, not the full probability distribution
 3. **No threshold behavior** - Can't answer "what happens if we stay under 2°C vs go to 2.1°C?"
 4. **Questionable at high temperatures** - The quadratic form breaks down above 4°C
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How Much Do Tipping Points Matter for Policy?

The 1% adjustment at 3°C might sound small, but it's actually quite significant:

Effect on the Social Cost of Carbon:

- **Without tipping points:** SCC would be about \$55/tCO₂ (rough estimate)
- **With tipping points:** SCC is \$66/tCO₂ in the baseline
- That's a **20% increase** in the carbon price

Effect on Optimal Temperature Target:

- Including tipping points **lowers** the optimal temperature from where it would otherwise be
- Makes the 2°C target look more economically justified
- The 1% extra damage at 3°C becomes 1.75% at 4°C, 2.75% at 5°C (scaling quadratically)

Effect on Net Present Value:

- The optimal climate policy provides **\$120 trillion** in benefits
- Without the tipping points adjustment, that number would be substantially lower
- Tipping points add roughly **\$30-40 trillion** to the economic case for action

The Bottom Line

DICE-2023's approach to tipping points:

- Takes state-of-the-art research (Dietz et al. 2021)
- Extracts the expected economic impact (1% GDP loss at 3°C)
- Incorporates it as a smooth adjustment to the damage function
- Quadratic form means it scales with temperature squared

What this captures: ✓ The expected economic cost of tipping points ✓ The general principle that risk increases with temperature ✓ Peer-reviewed quantification rather than speculation

What this misses: ✗ Threshold effects and sudden changes ✗ Irreversibility and hysteresis ✗ Probability distributions and fat tails ✗ Dynamic climate feedbacks from tipping ✗ Individual timing of different tips

Policy implication: The inclusion of tipping points makes climate action look significantly more valuable. But the smooth quadratic form probably **understates** the true risk, especially the small probability of really catastrophic outcomes. The model's tipping point treatment is **better than nothing but far from perfect**.

If You Want to Go Deeper

The actual Dietz et al. (2021) paper is much more sophisticated than what DICE-2023 uses. They found:

- **Best estimate:** Tipping points add 25% to smooth climate damages
- **With uncertainty:** Could be 10% to 50% depending on which tips occur
- **At higher temperatures:** The percentage increases dramatically
- **"Fat tail" risk:** Small chance of multiple simultaneous tips → economic catastrophe

DICE-2023 takes their central estimate and says "good enough for a tractable optimization model." Whether that's reasonable depends on your tolerance for catastrophic risk versus your need for a model that can actually be solved and optimized.

The philosophical question: Should we use expected values for potentially civilization-ending risks? Or should extreme tail events get special treatment? DICE-2023 says expected values are fine. Many climate economists disagree.

Citation

The tipping points component comes from:

Dietz, S., Rising, J., Stoerk, T., & Wagner, G. (2021). "Economic impacts of tipping points in the climate system." *Proceedings of the National Academy of Sciences*, 118(34), e2103081118.

How DICE-2023 incorporates it:

Barrage, L., & Nordhaus, W. (2024). "Policies, projections, and the social cost of carbon: Results from the DICE-2023 model." *Proceedings of the National Academy of Sciences*, 121(13), e2312030121.

The key quote from the DICE-2023 paper:

"The second component, based on a comprehensive study of tipping points (Dietz et al., 2021), adds a 1% output loss at a 3°C warming."